Expert Analysis: Entanglement Entropy Density as a Gravitational Source

Theoretical Tightening, OP3 Lindblad Model, NISQ Experiment Design, and Referee-Level Critique

Analysis prepared by: Expert Theoretical Physics Review

Responding to: *Refined Entanglement Gravity White Paper*, K. Monette (March 2026)

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Framework Summary (2–3 paragraphs as requested)

The refined white paper proposes adding a stress-energy source term $T^{\rm ent}_{\mu\nu} = \tilde{\kappa}(c^4/8\pi G k_{\rm B}\ln 2)S_{\rm ent}g_{\mu\nu}$ to the Einstein field equations, where $S_{\rm ent}$ is the entanglement entropy density of a quantum system in $\mathrm{bit\,m^{-3}}$ and $\tilde{\kappa}$ is a dimensionless coupling. The dimensional consistency of this addition is established regardless of the value of $\tilde{\kappa}$. The specific estimate $\tilde{\kappa} = -1/4$ is derived under a Planck-scale holographic regulator hypothesis — the ansatz that volume-law entanglement entropy in laboratory systems couples to spacetime geometry through the same Planck-length regulator that governs horizon area entropy. This is explicitly labeled a motivated hypothesis, not a theorem.

Three structural results are established correctly. First, the radial acceleration correction $\Delta a(R) \propto M_{\rm ent}(< R)/R^2$ for any bounded, spherically symmetric source — this resolves the prior confusion about $1/R$ versus $1/R^2$ scaling. Second, the Bianchi identity forces $\nabla^\mu T^{\rm matter}_{\mu\nu} = -\nabla_\nu(\tilde{\kappa}c^4 S_{\rm ent}/8\pi G k_{\rm B}\ln 2)$, so matter is not independently conserved in entanglement-entropy gradients; the paper adopts an "entanglement-sector exchange" interpretation where the *total* stress-energy is conserved. Third, existing equivalence-principle experiments (MICROSCOPE, Asenbaum et al.) do not directly constrain $\tilde{\kappa}$ because their test masses have negligible and nearly identical $S_{\rm ent}$.

The experimental program has been correctly revised: macroscopic ^{87}Rb BEC interferometry is demoted because Gaussian spatial screening gives $\alpha_{\rm screen} \sim e^{-(R/\xi)^2} \sim 10^{-43}$ for cloud radius $R \gg$ coherence length ξ, making the signal undetectable. The primary near-term target is now a 50–100 qubit NISQ platform in the regime $R \lesssim \xi$. The falsification criterion — requiring conditions F1–F5 on qubit count, pressure sensitivity, run count, platform independence, and null result — is stated without requiring knowledge of $\alpha_{\rm screen}$ or the Planck-scale renormalization of $\tilde{\kappa}$.

1 Tightening the Theoretical Core

1.1 Covariant notation and the full modified field equations

The white paper's modified field equations are restated here in the most standard covariant notation. Greek indices $\mu, \nu, \ldots$ run $0, 1, 2, 3$; signature $(-,+,+,+)$; $G = 6.674 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$; $c = 2.998 \times 10^8\,\mathrm{m\,s^{-1}}$; $k_{\rm B} = 1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$; $\ell_P = (\hbar G/c^3)^{1/2} = 1.616 \times 10^{-35}\,\mathrm{m}$.

Total stress-energy decomposition.

$$T^{\rm total}_{\mu\nu} \;\equiv\; T^{\rm matter}_{\mu\nu} \;+\; T^{\rm ent}_{\mu\nu}, \tag{1}$$

where

$$T^{\rm ent}_{\mu\nu} \;\equiv\; -\Lambda_{\rm ent}\, g_{\mu\nu}, \quad \Lambda_{\rm ent}(x) \;\equiv\; -\,\tilde{\kappa}\,\frac{c^4}{8\pi G k_{\rm B}\ln 2}\,S_{\rm ent}(x). \tag{2}$$

The sign convention: $\tilde{\kappa} < 0$ gives $\Lambda_{\rm ent} > 0$ (positive cosmological-constant-like term, repulsive curvature).

Modified Einstein equations.

$$G_{\mu\nu} \;\equiv\; R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} \;=\; \frac{8\pi G}{c^4}\,T^{\rm total}_{\mu\nu} \;=\; \frac{8\pi G}{c^4}\big(T^{\rm matter}_{\mu\nu} - \Lambda_{\rm ent}\,g_{\mu\nu}\big). \tag{3}$$

Contracted Bianchi identity. The second contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$ forces:

$$\nabla^\mu T^{\text{total}}_{\mu\nu} = 0 \implies \nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu} = \nabla_\nu \Lambda_{\text{ent}} = \tilde{\kappa}\, \frac{c^4}{8\pi G k_{\text{B}} \ln 2}\, \nabla_\nu S_{\text{ent}}. \tag{4}$$

This is an *exact* consequence of the field equations (3). It is not an additional assumption. It forces the interpretation adopted in the white paper: the right-hand side of Eq. (4) is the energy-momentum current flowing from the entanglement sector into the matter sector (or vice versa, depending on the sign of $\nabla_\nu S_{\text{ent}}$).

Expert Flag. Equation (4) implies a physical prediction that the white paper has not yet confronted explicitly. If matter is a point mass m at rest, its equation of motion in this background is

$$m\, \frac{Du^\mu}{d\tau} = \tilde{\kappa}\, \frac{c^4}{8\pi G k_{\text{B}} \ln 2}\, \frac{\nabla^\mu S_{\text{ent}}}{T^{\text{matter}}_{00}/c^2}, \tag{5}$$

where $D/d\tau$ is the covariant proper-time derivative. This is a classical fifth force proportional to ∇S_{ent} and independent of composition. The atom interferometry bound $\eta < 10^{-12}$ [3] bounds *composition-dependent* forces, but if two test masses are placed in regions of different ∇S_{ent}, Eq. (5) predicts different accelerations even without a new force law. This observable is not ruled out by current experiments and should be stated explicitly in the paper.

1.2 Holographic regulator steps: every assumption labeled

The estimate $\tilde{\kappa} = -1/4$ requires two assumptions beyond Jacobson's theorem. Both are labeled here with **[HRH]** (Holographic Regulator Hypothesis):

Step 1 **[Jacobson, established]** Apply $\delta Q = T_U\, dS_{BH}$ to a local Rindler horizon with $T_U = \hbar a/(2\pi c k_{\text{B}})$ and $dS_{BH} = (k_{\text{B}} c^3/4G\hbar)\, dA$. This gives $G_{\mu\nu} = 8\pi G c^{-4} T^{\text{matter}}_{\mu\nu}$.

Step 2 **[HRH-1]** For a volume with entanglement entropy density S_{ent} (bit m^{-3}), propose an additional entropy contribution to the horizon element dA:

$$dS_{\text{ent}} = \frac{S_{\text{ent}}}{k_{\text{B}}} \cdot \frac{\ell_P\, dA}{4\ell_P} = \frac{S_{\text{ent}}}{4k_{\text{B}}}\, dA. \tag{6}$$

Assumption: the volume element behind the horizon is exactly one Planck length deep ($dV = \ell_P\, dA$), so the full volume entropy density collapses to an area density via $S_{\text{ent}}\, dV \to S_{\text{ent}}\, \ell_P\, dA$. This regulation has no rigorous justification for non-horizon laboratory volumes. It is valid at causal horizons (where Bekenstein's bound enforces $S \leq A/4\ell_P^2$) but not for bulk systems with volume-law entanglement.

Step 3 **[HRH-2]** Add $T_U\, dS_{\text{ent}}$ to the Clausius heat flux. Following Jacobson's identification of heat with stress-energy flux gives:

$$T^{\text{ent}}_{\mu\nu} = -\frac{c^4}{32\pi G} S_{\text{ent}}\, g_{\mu\nu}, \tag{7}$$

which, matched to Eq. (2), yields $\tilde{\kappa} = -1/4$.

Assumption (implicit): The Clausius identification $\delta Q \leftrightarrow T^{\text{eff}}_{\mu\nu} k^\mu d\Sigma^\nu$ extends to the entanglement contribution. This requires that the entanglement entropy gradient contributes thermodynamically to the local horizon in the same way as matter energy-momentum. No proof exists.

The result $\tilde{\kappa} = -1/4$ is therefore contingent on both HRH-1 and HRH-2. Removing either assumption leaves the coupling as a free parameter.

1.3 Minimal addition to the source term for a natural conservation story

The current source term $T^{\text{ent}}_{\mu\nu} = -\Lambda_{\text{ent}}(x) g_{\mu\nu}$ is the simplest possible form (cosmological-constant structure with a varying prefactor). Its conservation equation (4) forces energy-momentum exchange with matter, which is physically plausible but lacks an action principle.

Proposed minimal extension: kinetic scalar-field term.

Promote S_{ent} to a dynamical scalar field ϕ and write:

$$T^{\text{ent}}_{\mu\nu} \;=\; \alpha_\phi\big(\nabla_\mu\phi\,\nabla_\nu\phi - \tfrac{1}{2} g_{\mu\nu}(\nabla\phi)^2\big) - \Lambda_\phi(\phi)\, g_{\mu\nu}, \tag{8}$$

where α_ϕ is a positive constant and $\Lambda_\phi(\phi) = -\tilde{\kappa}(c^4/8\pi G k_{\text{B}} \ln 2)\phi$. This has a natural action:

$$S_{\text{ent}}[\phi, g] = \int d^4x\, \sqrt{-g}\, \left[-\frac{\alpha_\phi}{2}(\nabla^\mu\phi\,\nabla_\mu\phi) + \Lambda_\phi(\phi)\right]. \tag{9}$$

Varying with respect to ϕ gives the scalar field equation of motion:

$$\alpha_\phi\,\Box\phi + \frac{d\Lambda_\phi}{d\phi} = \alpha_\phi\,\Box\phi + \tilde{\kappa}\frac{c^4}{8\pi G k_{\text{B}} \ln 2} = 0, \tag{10}$$

which relates the propagation of ϕ to the entanglement source density. Varying with respect to $g^{\mu\nu}$ gives Eq. (8) as the entanglement stress-energy. The *combined* system $(g_{\mu\nu}, \phi)$ satisfies $\nabla^\mu T^{\text{total}}_{\mu\nu} = 0$ automatically from the action principle.

Physical interpretation. In the limit $\alpha_\phi \to 0$, $\phi = S_{\text{ent}}$ is fixed by external conditions (the quantum state of the system), and Eq. (8) reduces to the white paper's current ansatz. For $\alpha_\phi > 0$, ϕ propagates as a massless scalar and can exchange energy-momentum with matter via the coupling in Eq. (4) without violating any conservation law.

Remark 1.1. The scalar ϕ is not a new fundamental field. It is an effective field representing the coarse-grained entanglement entropy density of the quantum system — an emergent degree of freedom, in the same spirit that thermodynamic quantities are emergent from microphysics. The action (9) should be understood as an effective field theory valid below the Planck scale, not a UV-complete theory. Whether ϕ can be derived from a fundamental quantum gravity theory (e.g., via holographic renormalization group) is Open Problem 10.

Conservation in the extended theory. With the action (9), the stress-energy of the full system satisfies:

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu}, \tag{11}$$

$$\nabla^\mu T^{\text{ent}}_{\mu\nu} = -\alpha_\phi\,(\Box\phi)\,\nabla_\nu\phi + \nabla_\nu\Lambda_\phi = \alpha_\phi(\alpha_\phi^{-1}\nabla_\nu\Lambda_\phi)\cdot(-\Box\phi) + \nabla_\nu\Lambda_\phi, \tag{12}$$

which vanishes on-shell (when ϕ satisfies its equation of motion (10)). This is the correct, covariant conservation story.

2 OP3: A Concrete Lindblad Model for Screening

2.1 The precise definition of S_{ent} that must be used

Before writing any Lindblad model, the white paper must specify exactly which entropy enters $T^{\text{ent}}_{\mu\nu}$. There are three candidates and they have *opposite* behaviors under decoherence:

1. **Total von Neumann entropy of the quantum system:** $S_{\text{total}} = -k_{\text{B}}\,\text{Tr}(\rho\ln\rho)$. Under decoherence, this *increases* (purity is lost). Using this would make decoherence *increase* the gravitational source — physically unnatural and inconsistent with the intuition that entanglement sources the effect.

2. **System-environment entanglement entropy:** $S_{\text{SE}} = -k_{\text{B}}\,\text{Tr}(\rho_S\ln\rho_S)$ where $\rho_S = \text{Tr}_E(\rho_{SE})$. This also increases under decoherence. Same problem.

3. **Internal (inter-qubit) entanglement entropy:** The von Neumann entropy of the reduced state of a subsystem traced over the other subsystems of the *same* quantum device, treating the environment as external. Concretely, for a bipartition $A|B$ of the device: $S_{\text{int}} = -k_{\text{B}}\,\text{Tr}(\rho_A\ln\rho_A)$ where $\rho_A = \text{Tr}_B\,\rho_{\text{device}}$. Under decoherence, $\rho_{\text{device}} \to$ a nearly classically correlated state, and $S_{\text{int}} \to 0$ (the system becomes product after tracing). This *decreases* under decoherence and is what the white paper implicitly intends.

Expert Flag. The white paper has not stated which of the three candidates it uses. Candidate (3) is the physically correct choice — it is the quantity that (a) decreases under decoherence, (b) is zero for a fully decohered state, and (c) is what experimentalists can verify via entanglement witnesses. All subsequent analysis in OP3 uses $S_{\text{ent}} = S_{\text{int}}$: the bipartite (or multi-partite) entanglement entropy of the internal degrees of freedom of the quantum device, tracing over the environment. This definition must be added explicitly to the white paper's Section 2.

2.2 Lindblad model for 50-qubit superconducting transmon qubits

Platform choice. We model a chain of $N = 50$ fixed-frequency transmon qubits, which is the architecture currently operated by IBM Quantum and Google. This choice is made for concreteness; the hierarchy developed below applies qualitatively to all NISQ platforms.

System Hamiltonian. In the rotating frame, in the two-level (qubit) approximation:

$$H_{\text{sys}} = \sum_{k=1}^{N} \frac{\omega_k}{2} \sigma_k^z + \sum_{\langle j,k \rangle} J_{jk} \left(\sigma_j^+ \sigma_k^- + \text{h.c.} \right), \tag{13}$$

where $\omega_k/2\pi \in [4.5, 5.0]\,\text{GHz}$ are qubit frequencies and $J_{jk}/2\pi \sim 2$–$10\,\text{MHz}$ are nearest-neighbor exchange couplings mediated by coplanar microwave resonators.

Lindblad operators and rates. The dominant environmental channels are:

(i) *Energy relaxation* (T1 process): $L_{1,k} = \sqrt{\gamma_1}\,\sigma_k^-$, rate $\gamma_1 = 1/T_1$. State of the art: $T_1 \sim 100$–$500\,\mu\text{s}$ (Google/IBM, 2025).

(ii) *Pure dephasing* (T2 process): $L_{\phi,k} = \sqrt{\gamma_\phi/2}\,\sigma_k^z$, rate $\gamma_\phi = 1/T_\phi$ where $T_\phi^{-1} = T_2^{-1} - T_1^{-1}/2$. State of the art: $T_2 \sim 100\,\mu\text{s}$, giving $T_\phi \sim 200\,\mu\text{s}$.

(iii) *Thermal excitation* (small at $T_{\text{fridge}} \sim 15\,\text{mK}$): $L_{\text{exc},k} = \sqrt{\bar{n}\gamma_1}\,\sigma_k^+$, where $\bar{n} = (e^{\hbar\omega_k/k_\text{B}T_{\text{fridge}}} - 1)^{-1} \lesssim 10^{-4}$. Negligible.

The master equation is:

$$\dot{\rho} = -\frac{i}{\hbar}[H_{\text{sys}}, \rho] + \sum_{k=1}^{N} \sum_{\alpha \in \{1,\phi\}} \gamma_\alpha \Big(L_{\alpha,k}\, \rho\, L_{\alpha,k}^\dagger - \tfrac{1}{2}\{L_{\alpha,k}^\dagger L_{\alpha,k}, \rho\} \Big). \tag{14}$$

2.3 Which correlators couple to the gravitational source

The gravitational source $S_{\text{ent}} = S_{\text{int}}$ depends on the full reduced density matrix $\rho_S(t) = \text{Tr}_E\, \rho_{SE}(t)$. For computational purposes, the most tractable proxy is the *single-site entropy*:

$$s_k(t) = -k_\text{B}\, \text{Tr}_k[\rho_k(t) \ln \rho_k(t)], \quad \rho_k = \text{Tr}_{\neq k}\, \rho_S, \tag{15}$$

where $\text{Tr}_{\neq k}$ traces over all qubits except qubit k. For a pure state, s_k is the entanglement entropy between qubit k and the rest.

For a GHZ state $|\text{GHZ}\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$, $s_k = k_\text{B} \ln 2$ for every k. Under qubit k's dephasing at rate γ_ϕ:

$$s_k(t) = k_\text{B}\, h\left(\frac{1 + e^{-\gamma_\phi t}}{2} \right), \tag{16}$$

where $h(p) = -p \ln p - (1-p)\ln(1-p)$ is binary entropy. However, a single-qubit dephasing event *also* collapses the GHZ entanglement between all other qubits: the GHZ correlation $|\langle 0^{\otimes N}|\rho|1^{\otimes N}\rangle|$ decays as $e^{-\gamma_\phi t}$ under a *single-qubit* dephasing with rate γ_ϕ, but under *any-qubit* dephasing the full GHZ coherence decays at rate $N\gamma_\phi$.

Therefore, the integrated entanglement entropy for GHZ during a measurement protocol of duration τ:

$$M_{\text{ent}}^{\text{GHZ}} \approx N k_\text{B} \ln 2 \int_0^\tau e^{-N\gamma_\phi t}\, dt = \frac{k_\text{B} \ln 2}{\gamma_\phi} \left(1 - e^{-N\gamma_\phi \tau} \right). \tag{17}$$

Maximized at $\tau \to \infty$: $M_{\text{ent}\,\text{GHZ},\max} = k_\text{B} \ln 2/\gamma_\phi$, independent of N (the GHZ state's total entanglement is lost on the timescale $1/(N\gamma_\phi)$, which decreases with N).

For a **cluster state**, which has short-range entanglement (each qubit entangled with its neighbors), the situation is more favorable. Under single-qubit dephasing:

$$M_{\text{ent cluster}} \approx N k_{\text{B}} \ln 2 \int_0^\tau e^{-\gamma_\phi t}\, dt = \frac{N k_{\text{B}} \ln 2}{\gamma_\phi} \left(1 - e^{-\gamma_\phi \tau}\right). \tag{18}$$

The cluster state's total entanglement grows with N because its local entanglement is protected from global collapse. This suggests cluster states are superior to GHZ states for maximizing M_{ent} under realistic decoherence.

Key finding: The choice of entangled state significantly affects the effective M_{ent}, and thus α_{screen}:

$$\frac{M_{\text{ent cluster}}}{M_{\text{ent}}^{\text{GHZ}}} \approx N \left(1 - e^{-\gamma_\phi \tau}\right) \gg 1 \quad \text{for } N \gg 1. \tag{19}$$

The experiment should use cluster states (or matrix product states with finite correlation length), not GHZ states, for maximum signal.

2.4 Hierarchy of regimes: qualitative expectations for α_{screen}

We define the effective screening factor as the ratio of actual to ideal integrated entanglement entropy:

$$\alpha_{\text{screen}} \equiv \frac{M_{\text{ent}}^{\text{actual}}}{M_{\text{ent}}^{\text{ideal}}}. \tag{20}$$

Regime I: Few-qubit ($N \lesssim 20$), short circuits ($d \lesssim 10$ gate layers).

- Gate fidelity: $F_{2\text{Q}} \geq 99.5\%$; circuit error per qubit $\varepsilon \lesssim 0.02$
- Coherence: $T_2 \gg N\tau_{\text{gate}}$; negligible decoherence during circuit
- Entanglement depth: achievable up to $k = N$ (full entanglement depth)
- α_{screen} estimate: $0.1 \lesssim \alpha_{\text{screen}} \lesssim 0.5$ (significant but not catastrophic suppression)
- M_{ent}: small (few qubits), signal correspondingly small

Regime II: Mesoscopic ($N = 50$–100), moderate circuits ($d \sim 20$–50 layers).

- Cluster state preparation requires $d \sim N/2 \sim 25$–50 CZ gate layers
- Decoherence per layer: $\varepsilon \sim N\gamma_\phi\tau_{\text{gate}} \sim 50 \times 10^4 \times 10^{-7} \approx 5 \times 10^{-3}$ per qubit per layer
- Total entanglement decay: cluster state entanglement decays at $\sim d\varepsilon \sim 0.1$–$0.25$ (10–25% loss per circuit)
- α_{screen} estimate (cluster state): $e^{-d\varepsilon} \sim 10^{-1}$ to 10^{-2}
- α_{screen} estimate (GHZ state): $e^{-N\gamma_\phi\tau} \sim 10^{-3}$ to 10^{-5}
- This is the **optimal regime**: large enough M_{ent}, small enough suppression

Regime III: Macroscopic ($N \gtrsim 10^3$), deep circuits or long coherence time.

- GHZ-type global entanglement: $\alpha_{\text{screen}} \sim e^{-N\gamma_\phi \tau} \to 0$ rapidly
- Local entanglement (cluster, MPS): α_{screen} may be $\mathcal{O}(0.01)$ but M_{ent}/N^α saturates as correlation length limits
- Macroscopic BEC: $R/\xi \sim 10$–100, Gaussian screening gives $e^{-(R/\xi)^2} \sim e^{-100}$ to e^{-10000}: effectively zero
- Verdict: not a viable near-term target for NISQ-based entanglement-gravity search

Open Problem. **OP3 Lindblad milestone path.**

1. *Month 1–2*: For a 50-qubit transmon chain with Hamiltonian (13) and Lindblad operators above, numerically solve (14) for $N = 5$, 10, 20 qubits with cluster state initial conditions. Compute $s_k(t)$ and $M_{\text{ent}}(t)$ as functions of $\gamma_\phi \tau$. This produces a calibrated $\alpha_{\text{screen}}(\gamma_\phi, N, \tau)$ function for the cluster state.
2. *Month 2–4*: Extend the Lindblad model to include gate errors (coherent and incoherent) using gate set tomography [6] data from actual hardware. Replace ideal unitary gates with their Kraus-map representations. Recompute M_{ent} including gate-level decoherence.
3. *Month 4–6*: Scale the numerical solution to $N = 50$ using tensor network (MPO/TDVP) methods. Identify the circuit depth d^* that maximizes $M_{\text{ent}}^{\text{total}} = \int_0^\tau \sum_k s_k(t)\, dt$.
4. *Month 6–12*: Analytically bound α_{screen} using the quantum data processing inequality and the monotonicity of relative entropy under CPTP maps. Establish whether α_{screen} is bounded away from zero for any circuit on any platform, or whether it necessarily vanishes in the thermodynamic limit.

The outcome of step 4 is the most important: if $\alpha_{\text{screen}} \to 0$ for all $N \to \infty$, the framework's experimental program must be confined to the few-qubit regime.

3 NISQ Experiment: Publishable Proposal

3.1 Platform: 2D reconfigurable neutral-atom array

Architecture. A $6 \times 6 = 36$ qubit array of ^{87}Rb atoms held in optical tweezer traps at lattice spacing $a = 5\,\mu$m. Two-qubit gates via Rydberg blockade using $|r\rangle = |n{=}70, S_{1/2}\rangle$ (blockade radius $r_b \approx 8\,\mu$m for nearest-neighbor coupling, $r_b/a \approx 1.6$, ensuring blockade on neighbors only).

Hilbert space. Qubit basis: $|0\rangle = |5S_{1/2}, F = 1, m_F = 0\rangle$, $|1\rangle = |5S_{1/2}, F = 2, m_F = 0\rangle$ (clock transition at $\omega_{hf}/2\pi = 6.835\,\text{GHz}$). Full Hilbert space: $\mathcal{H} = (\mathbb{C}^2)^{\otimes 36}$.

Hamiltonian (rotating frame, Rydberg dressing on for entangling gates):

$$H = \sum_{k=1}^{36} \left(\frac{\Omega_k}{2} \sigma_k^x - \frac{\Delta_k}{2} \sigma_k^z \right) + \sum_{k<l} \frac{C_6}{|\mathbf{r}_k - \mathbf{r}_l|^6} \, \hat{n}_k^{(r)} \, \hat{n}_l^{(r)}, \tag{21}$$

where $C_6/h = 862\,\mathrm{GHz} \cdot \mu\mathrm{m}^6$ for the $|70S_{1/2}\rangle$ state, Ω_k and Δ_k are Rabi frequency and detuning from a global or site-addressed laser, and $\hat{n}_k^{(r)} = |r\rangle\langle r|_k$ is the Rydberg population operator. During gates, $\Omega_k/2\pi \approx 3\,\mathrm{MHz}$ and $\Delta_k = 0$.

Decoherence parameters.

- Rydberg lifetime: $\tau_r \approx 0.3\,\mathrm{ms}$ at $T_{\mathrm{atom}} \sim 5\,\mu\mathrm{K}$
- CZ gate time: $\tau_{\mathrm{CZ}} \approx 200\,\mathrm{ns}$ (limited by $1/\Omega$)
- Single-qubit gate time: $\tau_{1\mathrm{Q}} \approx 500\,\mathrm{ns}$
- Coherence time (clock states): $T_2 > 1\,\mathrm{s}$ (magnetic-field-insensitive transition)
- Readout fidelity: $F_{\mathrm{read}} \approx 99.5\%$ (site-resolved fluorescence)

3.2 Branch A: High-entanglement states

State A1 (2D cluster state). Prepare via:

1. Initialize all qubits in $|+\rangle^{\otimes 36}$
2. Apply CZ_{kl} to all nearest-neighbor pairs (two layers: horizontal + vertical)
3. Result: 2D cluster state with entanglement depth $k = 36$ and bipartite entanglement $S_{A|B} \approx \min(|A|, |B|) \ln 2$ for any cut $A|B$

Circuit depth: 2 layers of CZ gates + 1 layer of Hadamards. Total time: $\tau_{\mathrm{prep}} \approx 2 \times 200\,\mathrm{ns} + 500\,\mathrm{ns} \approx 1\,\mu\mathrm{s}$. Entanglement entropy: $M_{\mathrm{ent}}^{\mathrm{tot}} \approx 36 k_{\mathrm{B}} \ln 2 \approx 25\, k_{\mathrm{B}}$.

State A2 (Checkerboard GHZ). Divide the 6×6 array into two sublattices (checkerboard pattern); prepare an 18-qubit GHZ state on each sublattice. Less fragile than full-array GHZ; $M_{\mathrm{ent}}^{\mathrm{tot}} \approx 2 k_{\mathrm{B}} \ln 2$.

3.3 Branch B: Low-entanglement reference states

State B1 (Product state, identical energy). Prepare $|+\rangle^{\otimes 36}$ with the same laser schedule as Branch A but with all CZ gates replaced by identity operations (zero-detuning pulses that return to the computational basis without entangling). Energy difference from A: at most $\sim N\langle J_{jk}\rangle \sim 36 \times 5\,\mathrm{MHz} \sim 10\,\mathrm{kHz}$ — must be characterized.

State B2 (Two-qubit entangled product). Prepare Bell pairs $|\Phi^+\rangle^{\otimes 18}$ (all 18 nearest-neighbor pairs entangled but no multipartite entanglement). $M_{\mathrm{ent}}^{\mathrm{tot}} \approx 18 k_{\mathrm{B}} \ln 2$ (different from A1 by a factor ~ 2, offering an intermediate comparison point).

3.4 Observable: Ramsey phase differential

Protocol.

1. Prepare Branch A (or B) state
2. Apply a global $\pi/2$ rotation around $\hat{x}$: maps $|0\rangle \to |+\rangle$, $|1\rangle \to |-\rangle$ on the clock transition
3. Free precession for time T_{Ramsey}
4. Apply a second $\pi/2$ rotation; measure population in $|1\rangle$
5. The accumulated phase $\phi = \omega_{\text{eff}}\, T_{\text{Ramsey}}$ encodes any shift in the effective hyperfine frequency

Signal structure. A coupling of the form $T^{\text{ent}}_{\mu\nu} = -\Lambda_{\text{ent}} g_{\mu\nu}$ modifies the local metric as $\delta g_{00} \approx -2\Phi_{\text{ent}}/c^2$ where Φ_{ent} is the Newtonian potential of the entanglement source. This produces a gravitational redshift of the hyperfine transition:

$$\frac{\delta\omega_{\text{hf}}}{\omega_{\text{hf}}} = \frac{\Phi_{\text{ent}}}{c^2} = -\frac{\tilde{\kappa}_{\text{eff}} c^2}{8\pi G k_{\text{B}} \ln 2} \frac{M_{\text{ent}}}{R^2_{\text{eff}}}, \tag{22}$$

where R_{eff} is the effective distance scale set by the qubit array geometry.

What this actually measures. The differential phase between Branch A and Branch B accumulated over T_{Ramsey}:

$$\Delta\phi_{AB} = (\omega^A_{\text{eff}} - \omega^B_{\text{eff}})\, T_{\text{Ramsey}} = \frac{\delta\omega^{AB}_{\text{hf}}}{\omega_{\text{hf}}}\, \omega_{\text{hf}}\, T_{\text{Ramsey}}. \tag{23}$$

This quantity bounds the combination

$$|\tilde{\kappa}_{\text{eff}}| \cdot \frac{M^A_{\text{ent}} - M^B_{\text{ent}}}{R^2_{\text{eff}}} < \frac{|\Delta\phi_{AB}|_{\text{sens}}}{T_{\text{Ramsey}}} \cdot \frac{8\pi G k_{\text{B}} \ln 2}{c^2\, \omega_{\text{hf}}}, \tag{24}$$

without requiring knowledge of α_{screen} or the Planck-scale renormalization. The right-hand side is a pure number set by the experimental sensitivity.

For current state-of-the-art Ramsey spectroscopy on Rb clock transitions: sensitivity $\delta(\delta\omega)/\omega \sim 10^{-14}$–$10^{-16}$ per shot in optimal conditions (limited by quantum projection noise at $\delta\phi \sim 1/\sqrt{N_{\text{atoms}}}$ per shot, and by measurement time). After 10^4 shots: $\delta(\Delta\phi) \sim 10^{-3}$ rad.

3.5 Systematic error budget and fatal vs. non-fatal errors

The fatal systematic in detail. The entangled state (A1, 2D cluster) has different two-body correlators $\langle \hat{n}_j \hat{n}_k \rangle$ than the product state (B1). For the cluster state, $\langle \hat{n}_j \hat{n}_k \rangle_{\text{cluster}} = 1/4$ for all pairs (since each qubit is in a $|+\rangle$ state locally). For $|+\rangle^{\otimes N}$, $\langle \hat{n}_j \hat{n}_k \rangle_{\text{product}} = 1/4$ as well. So this specific pair of states has *identical* two-body correlators. This is why cluster states are preferred over GHZ for this comparison: the differential mean-field energy is zero.

Table 1: Systematic error budget for the neutral-atom Ramsey protocol.

Systematic	Physical origin	Mitigation	Fatal?
Differential mean-field energy shift	$\sum_{kl} J_{kl}(\langle \hat{n}_k \hat{n}_l \rangle_A - \langle \hat{n}_k \hat{n}_l \rangle_B)$	Match two-body correlators between A and B (e.g., use B2 with same $\langle \hat{n}\hat{n} \rangle$)	**Yes** if uncontrolled
Differential AC Stark shift from gate pulses	Gate laser amplitude $\propto \Omega^2/\Delta_{\text{laser}}$	Zero-area pulse sequences; same laser schedule for A and B	No
Magnetic field fluctuations	$\partial B/\partial t \neq 0$	Clock transition insensitive to first order; active shielding	No
Differential trap displacement (A vs B)	Different motional state for entangled atoms	Common-mode rejection from differential measurement; same trapping geometry	No
SPAM errors in entanglement verification	Readout fidelity < 1	Entanglement witnesses with SPAM calibration (GST)	No
Atom loss during circuit	Rydberg decay, collision loss	Monitor atom number; postselect on full count	No (postselect)
Differential motional heating	Heating rate $\dot{n}_{\text{phonon}}$	Sideband cooling; matched motional temperature for A and B	No

However, for GHZ vs. product: $\langle \hat{n}_j \hat{n}_k \rangle_{\text{GHZ}} = 1/4$ but the GHZ has non-trivial N-body correlators that create different collective energy shifts. This must be verified numerically for the specific Hamiltonian.

Null-result interpretation. If $\Delta\phi_{AB} = 0$ at the level of experimental sensitivity, this rules out:

$$|\tilde{\kappa}_{\text{eff}}| \cdot \frac{\Delta M_{\text{ent}}}{R_{\text{eff}}^2} > \frac{|\Delta\phi|_{\text{sens}}\, 8\pi G k_{\text{B}} \ln 2}{c^2\, \omega_{\text{hf}}\, T_{\text{Ramsey}}}. \tag{25}$$

With $R_{\text{eff}} \sim 3a = 15\,\mu\text{m}$, $\Delta M_{\text{ent}} \sim 18 k_{\text{B}} \ln 2$, $T_{\text{Ramsey}} = 0.1\,\text{s}$, $|\Delta\phi|_{\text{sens}} \sim 10^{-3}\,\text{rad}$:

$|\tilde{\kappa}_{\text{eff}}| < \frac{10^{-3} \times 8\pi \times 6.67 \times 10^{-11} \times 9.57 \times 10^{-24}}{(3 \times 10^8)^2 \times 6.84 \times 10^9 \times 0.1} \times \frac{(1.5 \times 10^{-5})^2}{18 \times 9.57 \times 10^{-24}} \approx 10^{-47}$.

This is an extraordinarily small bound on $\tilde{\kappa}_{\text{eff}}$, confirming OP10: the coupling prefactor $c^4/(8\pi G k_{\text{B}} \ln 2)$ is so large that even with $\alpha_{\text{screen}} \sim 10^{-2}$, the experiment constrains $\tilde{\kappa}$ to be far below 10^{-15} — meaning the falsification criterion F1–F5 is much harder to reach than the white paper suggests at face value.

Expert Flag. The experimental bound on $|\tilde{\kappa}_{\text{eff}}|$ from the proposed NISQ experiment is approximately 10^{-47}, not the 10^{-15} stated in the falsification criterion. This reflects the enormous ratio $c^4/(8\pi G k_{\text{B}} \ln 2) \approx 5 \times 10^{65}$ in SI units. The gap between 10^{-47} and 10^{-15} (a factor of 10^{32}) is precisely OP10: if the Planck-scale renormalization suppresses $\tilde{\kappa}$ to $\sim 10^{-47}$ at laboratory scales, the framework is self-consistent and both bounds agree. If $\tilde{\kappa}$ is renormalized to a value larger than 10^{-47} (e.g., 10^{-15}), the proposed experiment has already falsified it. This should be stated explicitly in the paper's experimental section.

4 Mock Referee Report and Three-Paper Sequence

4.1 Mock referee report

Referee Report
Journal: Physical Review D (or Classical and Quantum Gravity)
Manuscript: Entanglement Entropy Density as a Gravitational Source

SUMMARY

The manuscript presents a phenomenological extension of Jacobson's thermodynamic gravity in which entanglement entropy density acts as an additional gravitational source term. The dimensional consistency of the modification is established, and the paper is admirably honest about where derivations end and hypotheses begin. The resolution of the radial scaling question (OP4) and the treatment of the Bianchi identity are genuine improvements over prior drafts.

MAJOR STRENGTHS

(1) *Epistemic clarity.* The separation between established results (dimensional consistency, radial scaling), motivated hypotheses (Planck regulator, $\tilde{\kappa} = -1/4$), and unresolved parameters (α_{screen}, renormalization) is one of the clearest in speculative gravity phenomenology literature. This alone makes the paper worth reading.

(2) *OP4 resolution.* The derivation $\Delta a(R) \propto M_{\text{ent}}(<R)/R^2$ from the modified Poisson equation is correct and removes a structural inconsistency from prior work.

(3) *Falsification orientation.* The F1–F5 criterion is quantitative and platform-independent. The paper correctly notes that null results are scientifically valuable even before α_{screen} is resolved.

MAJOR CONCERNS (revisions required)

(C1) *Definition of S_{ent} is missing.* The manuscript uses S_{ent} throughout but never states precisely which entropy it is. Three options exist (total von Neumann, system-environment, internal inter-qubit) with opposite behaviors under decoherence. The analysis requires candidate (3) (internal entanglement entropy that decreases under decoherence); this must be stated explicitly. Without it, the sign of α_{screen}'s effect on the gravitational source is ambiguous.

(C2) *The fifth-force prediction from the Bianchi identity is not confronted.* Equation (4) predicts a classical force on matter proportional to ∇S_{ent}. For a single qubit near an entan-

gled register, this force is not composition-dependent (unlike MICROSCOPE's constraint) but is entanglement-density-dependent. The paper must either (a) bound this force from existing precision measurements, or (b) state explicitly that it is an unconstrained prediction. Claiming MICROSCOPE does not constrain $\tilde{\kappa}$ is correct but incomplete: MICROSCOPE tests composition-dependence, not entanglement-dependence.

(C3) *The signal magnitude is inconsistent with the falsification criterion.* The falsification criterion (F2: $\Delta p < 10^{-6}\,\mathrm{Pa}$) and the proposed NISQ experiment produce a bound on $|\tilde{\kappa}_{\mathrm{eff}}|$ that is approximately 10^{-47}, not 10^{-15}. Either the falsification criterion should be tightened to match realistic experiments, or the criterion should be rephrased as bounding $|\tilde{\kappa}_{\mathrm{eff}}| \cdot M_{\mathrm{ent}}/R^2$ directly without implying anything about $\tilde{\kappa}$ alone.

(C4) *OP3 is not specific enough to constitute an open problem statement.* "Derive α_{screen} from a Lindblad master equation" is a research direction, not an open problem. A publishable open problem statement must specify: the platform, the specific correlators that couple to S_{ent}, the Lindblad operators, and what a "solution" would consist of. The milestone path outlined in Section 2.4 of this analysis is the kind of specificity required.

(C5) *The conservation interpretation (entanglement-sector exchange) is adopted without justification.* Of the three interpretations (fifth force, sector exchange, incomplete source term), interpretation (2) is chosen as "more natural" without argument. A natural justification requires an action principle, which the manuscript lacks. The paper should either (a) present the scalar-field effective action (Eq. (9)) as minimal supporting structure, or (b) explicitly state that the conservation story is an ansatz awaiting an action derivation.

SPECIFIC REQUIRED REVISIONS

(R1) Add explicit definition: $S_{\mathrm{ent}} = S_{\mathrm{int}}(x)$ = bipartite entanglement entropy density of the internal quantum degrees of freedom, tracing over the environment. Clarify that this decreases under decoherence.

(R2) Add a subsection (or boxed remark) working out the classical fifth-force prediction from Eq. (4) for a representative test case (e.g., a classical test mass near a superconducting chip), and quantify how it compares to known precision measurements.

(R3) Add a sentence to the falsification criterion noting that F2 ($\Delta p < 10^{-6}\,\mathrm{Pa}$) translates to $|\tilde{\kappa}_{\mathrm{eff}}| \cdot M_{\mathrm{ent}}/R^2 <$ [specific number from the proposed experiment's geometry], making the bound on $\tilde{\kappa}$ dependent on α_{screen} and the Planck-scale renormalization.

(R4) Either present the scalar-field action (9) as a minimal extension that makes the conservation story covariant and natural, or add a "model uncertainty" box noting that the conservation interpretation is unjustified without an action.

(R5) Concrete intermediate milestones for OP3: see Section 2.4. The current statement "derive α_{screen} from Lindblad" is too vague for a publishable open problem.

PUBLISHABILITY RECOMMENDATION

Reject with major revisions required. If C1–C5 are adequately addressed, the paper would represent a publishable contribution as a theory proposal paper in Physical Review D or Classical and Quantum Gravity. It should not be submitted to Physical Review Letters until OP3 is resolved (at least at the qualitative level in one platform).

4.2 Three-paper sequence

Based on the referee critique, the most defensible publication strategy is a *three-paper sequence*, each making a single, independently defensible claim.

Paper I: Theory Letter (4–6 pages, PRL/PRD Letters)

Claim: Dimensional consistency, $\tilde{\kappa}$ estimate under explicit regulator hypothesis, OP4 resolution, Bianchi-identity treatment with scalar-field extension, falsification criterion.

Target journal: Physical Review D (Letters format) or Physical Review Research.

Prerequisite: R1 (define S_{ent}), R4 (scalar-field action or explicit caveat), R3 (reformulate falsification criterion).

What it does NOT need: OP3 resolution, experimental result, any specific platform.

Paper II: Technical Theory Backup (20–30 pages, PRD)

Claim: Full derivation of OP4, complete OP3 Lindblad model for one platform (50-qubit transmon chain), regime hierarchy for α_{screen}, OP8 Bianchi analysis with quantitative bounds from equivalence-principle tests, OP10 renormalization group power-counting argument.

Target journal: Physical Review D (full article).

Prerequisite: Numerical solution of Lindblad equation for cluster states, $N \leq 20$ qubits. Tensor network extension to $N = 50$.

Required new work: Steps 1–2 of the OP3 milestone path (Section 2.4 above).

Paper III: Experimental Proposal (10–15 pages, PRA or npj Quantum Info.)

Claim: A concrete 36-qubit neutral-atom Ramsey experiment that (a) sets a bound on $|\tilde{\kappa}_{\text{eff}}| \cdot M_{\text{ent}}/R^2$ and (b) is publishable as a platform-agnostic experimental proposal even without performing the measurement.

Target journal: Physical Review A (experimental) or npj Quantum Information.

Content: Exact Hamiltonian and Lindblad parameters, Branch A/B state preparation protocols, Ramsey measurement design, complete systematic error budget (Table 1), and the null-result interpretation bound.

Required new work: The mean-field energy matching calculation for cluster vs. product states. The differential frequency shift from the local entanglement entropy density.

Timeline.

Months	Task	Paper
1–3	R1–R5 revisions to current white paper + scalar field action	I (theory letter)
3–6	Lindblad numerics, $N \leq 20$, cluster state α_{screen}	II (regime hierarchy section)
6–12	Tensor network extension to $N = 50$; OP3 complete	II (full)
12–18	Neutral atom experiment design + mean-field systematics	III (full)

Conclusion: What Is Solid, What Is Open, and What To Do Next

What is solid and should be published. The dimensional consistency of the modified field equation, the OP4 resolution ($\Delta a \propto M_{\text{ent}}/R^2$), the Bianchi identity consequences including

Eq. (4), the MICROSCOPE non-constraint argument, and the falsification criterion structure (F1–F5 rephrased as a bound on $|\tilde{\kappa}_{\rm eff}|M_{\rm ent}/R^2$). These are internally consistent results that can be published as Paper I now.

What is open but not fatal. The $\tilde{\kappa} = -1/4$ estimate. This is motivated and internally consistent. It cannot be treated as a prediction until HRH-1 and HRH-2 are independently justified. The paper correctly labels it as a hypothesis.

What is open and important to resolve. OP3 ($\alpha_{\rm screen}$ for real platforms): the milestone path in Section 2.4 provides a concrete plan requiring 3–6 months of numerical work. OP10 (Planck-scale renormalization): harder, requiring quantum gravity tools, but power-counting arguments in effective field theory may provide qualitative guidance within 6–12 months.

The single most important immediate action. Define $S_{\rm ent}$ precisely as the internal inter-qubit entanglement entropy density, verify that the cluster state maintains large $\alpha_{\rm screen}$ under realistic decoherence for $N = 50$ (using Regime II estimates), and restate the falsification criterion as a bound on the observable combination $|\tilde{\kappa}_{\rm eff}|M_{\rm ent}/R^2$. These three changes transform the paper from a work with significant ambiguities to one that is genuinely publishable.

References

[1] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260 (1995).

[2] E. Verlinde, "Emergent gravity and the dark universe," *SciPost Phys.* **2**, 016 (2017).

[3] P. Asenbaum, C. Overstreet, T. Kim, J. Curti, and M. A. Kasevich, "Atom-interferometric test of the equivalence principle at the 10^{-12} level," *Phys. Rev. Lett.* **125**, 191101 (2020).

[4] P. T. Touboul *et al.* (MICROSCOPE Collaboration), *Phys. Rev. Lett.* **129**, 121102 (2022).

[5] S. T. Flammia, D. Gross, Y.-K. Liu, and J. Eisert, *New J. Phys.* **14**, 095022 (2012).

[6] R. Blume-Kohout *et al.*, *Nat. Commun.* **8**, 14485 (2017).

[7] S. de Haro, S. N. Solodukhin, and K. Skenderis, "Holographic reconstruction of spacetime and renormalization in the AdS/CFT correspondence," *Commun. Math. Phys.* **217**, 595 (2001).

[8] H. Bernien *et al.*, "Probing many-body dynamics on a 51-atom quantum simulator," *Nature* **551**, 579 (2017).

[9] M. Saffman, "Quantum computing with atomic qubits and Rydberg interactions: Progress and challenges," *J. Phys. B* **49**, 202001 (2016).

[10] S. L. Braunstein and C. M. Caves, *Phys. Rev. Lett.* **72**, 3439 (1994).